



Natural Language Processing

BITS Pilani
Pilani Campus

Dr. Chetana Gavankar, Ph.D,
IIT Bombay-Monash University Australia
Chetana.gavankar@pilani.bits-pilani.ac.in



Session 1

These slides are prepared by the instructor, with grateful acknowledgement of James Allen and many others who made their course materials freely available online.

Session Content

- Objective of course
- What will we learn in this course?
- Text books and Reference books
- Evaluation Plan
- What is Natural Language Processing?
- Application areas of Natural Language Processing
- Introduction to Natural Language Processing



Objective of course

- To Identify and recall fundamental concepts and techniques in Natural Language Processing (NLP).
- To Explain the computational properties of natural languages and articulate the algorithms commonly utilized for processing linguistic information.
- To Apply basic mathematical models and methods in NLP applications to solve problems and perform tasks.
- To Examine research and development efforts in Natural Language Processing to discern trends, challenges, and opportunities.

What you will learn in this course

- **Vector Semantics**
- **Word Embedding**
- **Language Modelling**
 - N-gram language modeling
 - Neural Language Models
- **Introduction to LLM and Prompt Engineering**
- **Part-of-Speech Tagging**
- **Parsing**
- **Encoder-Decoder Models, Attention and Contextual Embedding's, Introduction to BERT, GPT**
- **Word sense disambiguation**
- **Semantic web ontology and knowledge Graphs**
- **Retrieval Augmented Generation (RAG)**
- **Text Summarization**

Text books and Reference books



T1	Jurafsky and Martin, SPEECH and LANGUAGE PROCESSING: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, McGraw Hill
T2	Manning and Schütze, Foundations of Statistical Natural Language Processing, MIT Press. Cambridge, MA
R1	Allen James, Natural Language Understanding
R2	Neural Machine Translation by Philipp Koehn
R3	Semantic Web Primer (Information Systems) By Antoniou, Grigoris; Van Harmelen, Frank

Evaluation Plan



Code	Name	Weight	Date
EC1	Quiz1	5% [Best of 2]	24 May 2026 to 29 May 2026
	Quiz2		26 July 2026 to 1st August 2026
	EL Assignment I	12%	23 May 2026 to 15 June 2026
	EL Assignment II	13%	1st Aug to 31st Aug
EC2	Midsem Exam [Regular, Make-Up]	30%	20-06-2026 (FN) 27-06-2026 (FN)
EC3	Comprehensive Exam [Regular, Make-Up]	40%	05-09-2026 (FN) 12-09-2026 (FN)



What is Natural Language Processing?

- Analyze, understand and generate human languages just like humans do.
- To explain linguistic theories, to use the theories to build systems that can be of social use..
- Started off as a branch of Artificial Intelligence..
- Borrows from Linguistics, Psycholinguistics, Cognitive Science & Statistics.
- Make computers learn our language rather than we learn theirs.

Why Study NLP?

- A hallmark of human intelligence.
- Text is the largest repository of human knowledge and is growing quickly.
 - emails, news articles, web pages, IM, scientific articles, insurance claims, customer complaint letters, transcripts of phone calls, technical documents, government documents, patent portfolios, court decisions, contracts,



Why Study NLP?

- Natural Language Processing (NLP) is experiencing explosive growth, driven by the rapid adoption, Large Language Models (LLMs), Agentic AI
- The global NLP market is projected to grow significantly, with some forecasts suggesting it will reach over \$140 billion by 2033.

Key roles include:

- **NLP Engineer:** Focuses on model design and training using frameworks like Hugging Face and TensorFlow.
- **Prompt Engineer:** Specializes in refining inputs for Large Language Models (LLMs).
- **Conversational AI Developer:** Builds chatbots and virtual assistants.
- **Data Scientist (NLP-focused):** Analyzes text data for insights and sentiment.
- **AI Platform Engineer:** Deploys and manages NLP models in production.
- **Computational Linguist:** Bridges linguistics and programming for text analysis.



Why are language technologies needed?

- Many companies make a lot of money if they could use computer programmes that understood text or speech.
 - answering the phone, and replying to a question
 - understanding the text on a Web page to decide who it might be of interest to
 - translating a daily newspaper from Japanese to English
 - understanding text in journals / books and building an expert systems based on that understanding

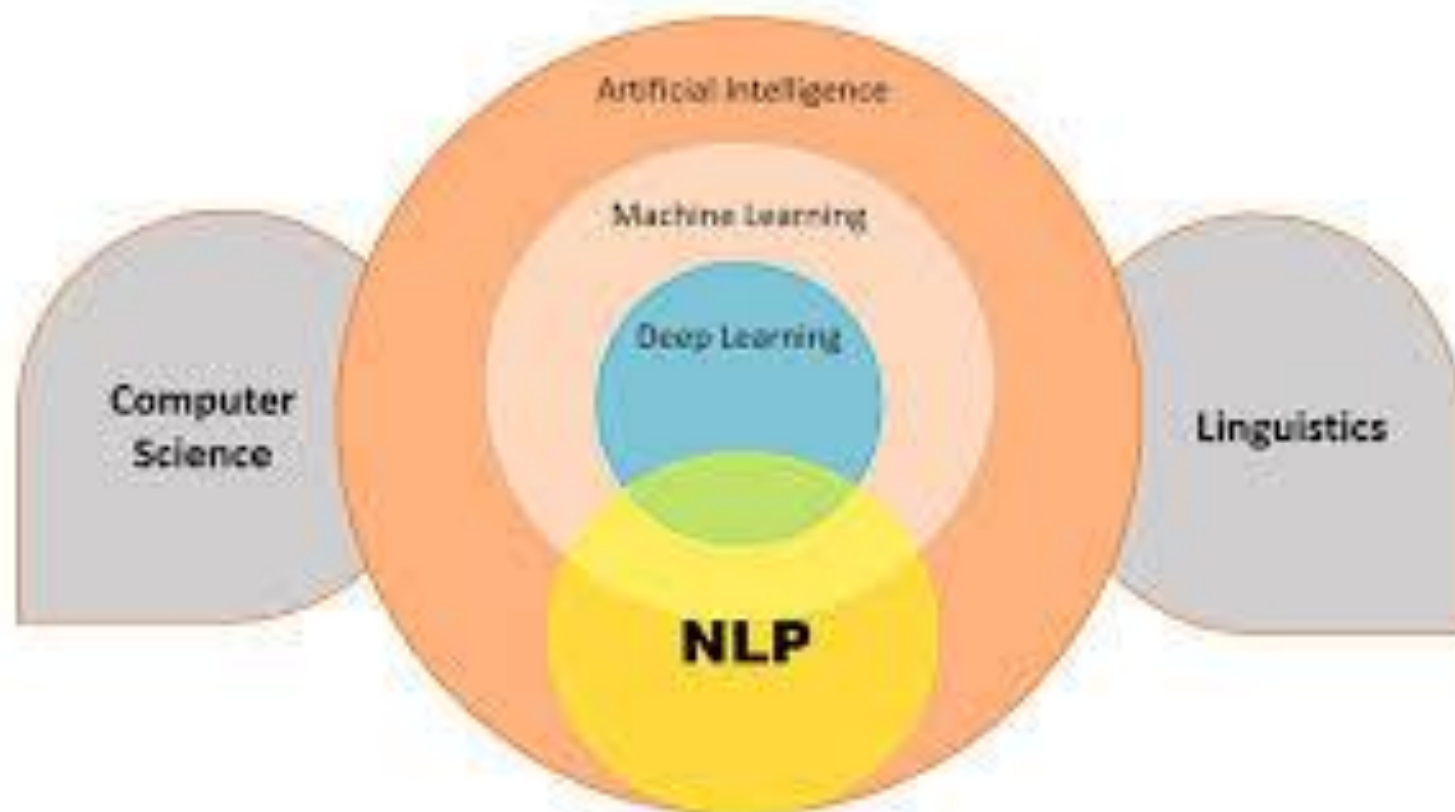
Dreams or reality??

- Will my computer talk to me like another human ??
- Will the search engine get me exactly what I am looking for??
- Can my PC read the whole newspaper and tell me the important news only..??
- Can my palmtop translate what that Japanese lady is telling me.. ??
- Ahhh.. Can my PC do my NLP assignments ??
- Do you know how our brain processes language ??

Dream come true?



Where does it NLP fit in AI and CS?



Brief history of NLP



Era / Period	Key Developments	Highlights / Examples
1950s–1960s (Early NLP)	Rule-based systems	Turing Test, early machine translation, ELIZA
1970s–1980s (Symbolic NLP)	Linguistic rules & grammars	Context-free grammars, expert systems, knowledge-based approaches
1990s (Statistical NLP)	Data-driven methods	N-grams, Hidden Markov Models, probabilistic parsing
2000s (Machine Learning NLP)	Supervised learning	SVMs, CRFs, feature engineering, rise of corpora like WordNet
2010–2017 (Deep Learning)	Neural networks dominate	RNNs, LSTMs, word embeddings like Word2Vec, GloVe
2018–2022 (Transformer Era)	Attention mechanisms	BERT, GPT, transfer learning
2023–Present (LLMs & Agentic AI)	Large-scale foundation models	ChatGPT, multimodal models, retrieval-augmented generation, autonomous agents

Introduction to Natural Language Processing



- The Study of Language
- Application areas
- Code Demo
- Why is NLP a big deal
- The Different Levels of Language understanding
- The Organization of Natural Language Processing Systems
- Evaluating NLP systems



Study of language:

Can machines speak our language?

Dave: Open the pod bay doors, HAL.

Hal: I am sorry, Dave. I am afraid I can't do that.

Dave: What's the problem.

Hal : I think you know what the problem is just as well as I do.

Dave: I don't know what you're talking about.

Hal : I know that you and Frank were planning to against me, and I'm afraid that's something I cannot allow to happen.

General speech and language understanding and generation capabilities

Politeness: emotional intelligence

Self-awareness: a model of self, including goals and plans

Belief ascription: modeling others; reasoning about their goals and plans

Study of language: Can machines speak our language?



Hal: I can tell from the tone of your voice, Dave, that you're upset.
Why don't you take a stress pill and get some rest.

[Dave has just drawn another sketch of Dr. Hunter].

Hal : Can you hold it a bit closer?

[Dave does so].

Hal : That's Dr. Hunter, isn't it?

Dave: Yes.

Recognition of emotion from speech

Vision capability including visual recognition of emotions and faces

Also: situational ambiguity

Study of language: Can machines speak our language?



To attain the levels of performance we attribute to Hal, we need to be able to define, model, acquire and manipulate

- Knowledge of the world and of agents in it,
- Text meaning,
- Intention

Application areas

- Text-to-Speech & Speech recognition
- Healthcare
- Natural Language Dialogue Interfaces to Databases
- Information Retrieval
- Information Extraction
- Document Classification
- Document Image Analysis
- Automatic Summarization (<https://quillbot.com/summarize>)
- Text Proof-reading – Spelling & Grammar_
- Machine Translation
- **Fake News and Cyberbullying Detection**
- **Monitoring Social Media Using NLP**
- Plagiarism detection
- Look-ahead typing / Word prediction_
- Question Answering System (<http://start.csail.mit.edu/index.php>)
- Sentiment Analysis (<https://komprehend.io/sentiment-analysis>)

Some commercial tools

- [Google Cloud Natural Language API](#): Provides strong pre-trained models for entity analysis, sentiment, and classification.
- [Amazon Comprehend](#): Specialized in finding insights and relationships in text, including PII detection.
- [Microsoft Azure Text Analytics](#): Excellent for enterprise-level applications, particularly in healthcare.
- [IBM Watson NLP](#): Used for advanced, industry-specific text understanding.

Open Source Tools

- [spaCy](#): Known for industrial-strength speed and efficiency in production pipelines.
- [NLTK \(Natural Language Toolkit\)](#): The standard for education, research, and quick prototyping in Python.
- [Hugging Face Transformers](#): The go-to library for state-of-the-art, pre-trained language models like BERT and GPT.
- [Stanford CoreNLP](#): A robust Java-based tool offering deep linguistic annotations.
- [TextBlob](#): A beginner-friendly library for common NLP tasks

- The Natural Language Toolkit (NLTK) is a platform used for building programs for text analysis.

Open Anaconda terminal, run

pip install nltk.

Anaconda and Jupiter are best and popular data science tools

In Jupyter, the console commands can be executed by the '!' sign before the command within the cell.

! pip install nltk

[NLTK book](#)

[NLTK discussion forum](#)

<https://www.nltk.org/install.html>

NLTK Demo

Cleaning (or pre-processing) the data typically consists of a number of steps:

- **Remove punctuation**
- **Tokenization**
- **Remove stop words**
- **Lemmatize/Stem**

Why is NLP Big Deal



- L = Words + rules + exceptions..
- Ambiguity at all levels..
- We speak different languages..
- And language is a cultural entity..
- So they are not equivalent..
- Highly systematic but also complex..
- Keeps changing.. New words, New rules and New exceptions..
- Source : Electronic texts / Printed texts / Acoustic Speech Signal.. they are noisy..
- Language looks obvious to us.. But it is a Big Deal 😊!



Why is NLP difficult?



Where is **Black Panther** playing in **Mountain View**?

Black Panther is playing at the Century 16 Theater.

When is **it** playing **there**?

It's playing at 2pm, 5pm, and 8pm.

OK. I'd like 1 **adult** and 2 **children** for **the first show**.
How much would **that** cost?



Need **domain knowledge**, **discourse knowledge**, **world knowledge**

Types of Ambiguities

I. Structural Ambiguities

- Namrata thinks she understands me.
- She thinks Namrata understands me.
- Visiting relatives can be nuisance. (two meanings)

II. Grammatical Ambiguities

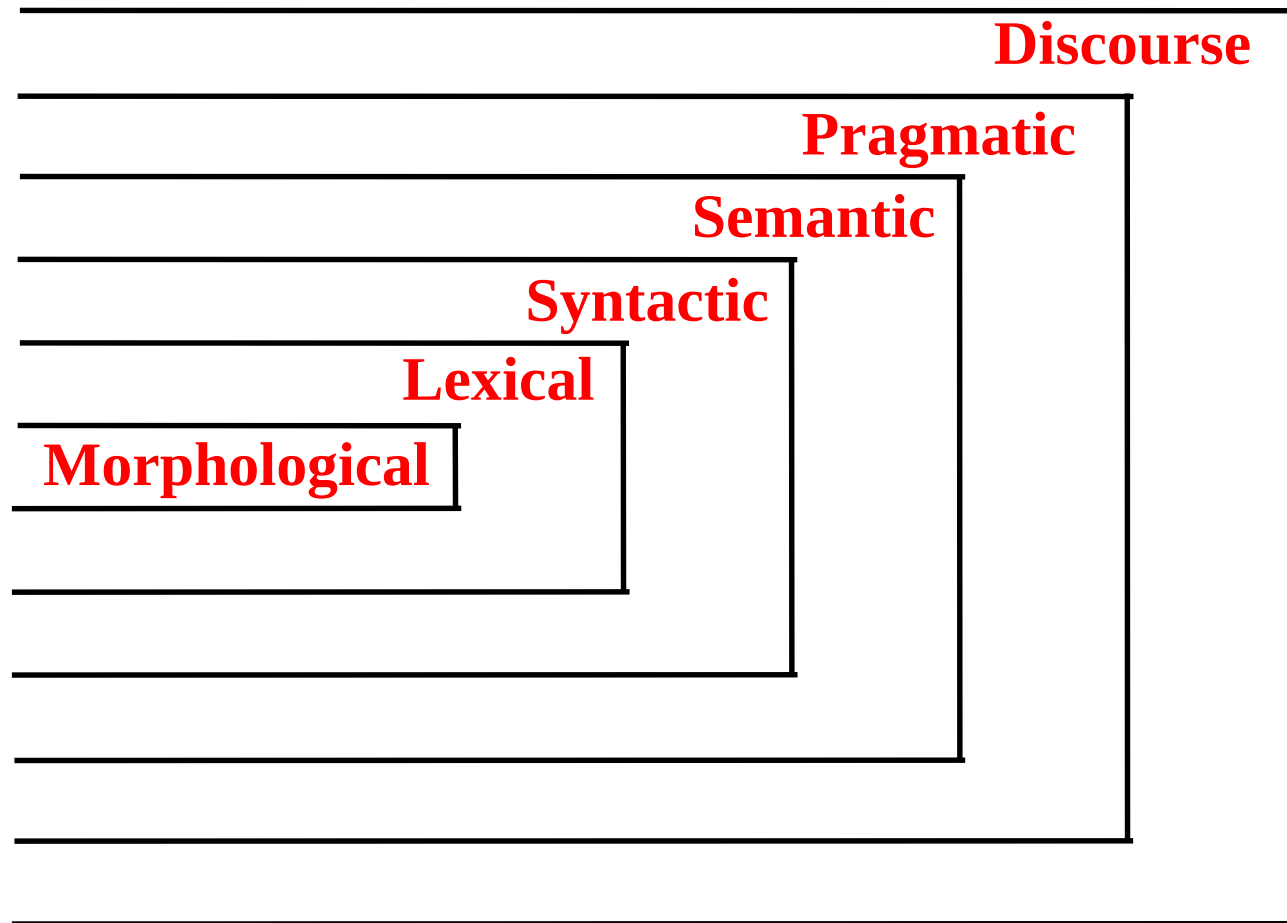
- I (feminine or masculine) go.
- Can- Noun = container, Can – Modal(auxiliary verb),
Can-verb = to can means to pack etc

III. Lexical Ambiguities:

Polysemy Ex: "understand" (I get it)

- Homonymy Ex: Bank= river, financial bank

Different Levels of Language Analysis





Levels of language understanding

Morphological Knowledge : Concerns how words are constructed from basic meaning units called morphemes. A morpheme is the primitive unit of meaning in a language (Ex: “*friendly*” is derived from the meaning of noun “*friend*” and suffix “*-ly*”, which transforms noun into adjective)

Lexical Knowledge : Concerns with listing of words and categorizing them Ex: *friendly* or *beautyship* is incorrect lexically. But *friendship* and *beautiful* is correct

Syntactic Knowledge : Concerns how words can be put together to form correct sentences and determines what structural role each word plays in the sentence and what phrases are subpart of other phrases Ex: “*Large have green ideas nose*” is lexically correct but syntactically incorrect.



Levels of language understanding

Semantic Knowledge :Concerns what words mean and how these meanings – combine in sentences to form sentence meanings. This is the study of context-independent meaning. Ex: “Green ideas have large noses” is syntactically correct but semantically incorrect.

Pragmatic Knowledge :Concerns how sentences are used in different situations how use affects the interpretation of sentence “She cuts banana with a pen” is semantically correct but pragmatically incorrect as it has no useful meaning.

Discourse Knowledge :Concerns how the immediately preceding sentences affect the interpretation of next sentence

Ex: Chetana completed PhD student at IIT Bombay. She is a Professor at BITS Pilani.

Levels of language understanding in Agentic AI era



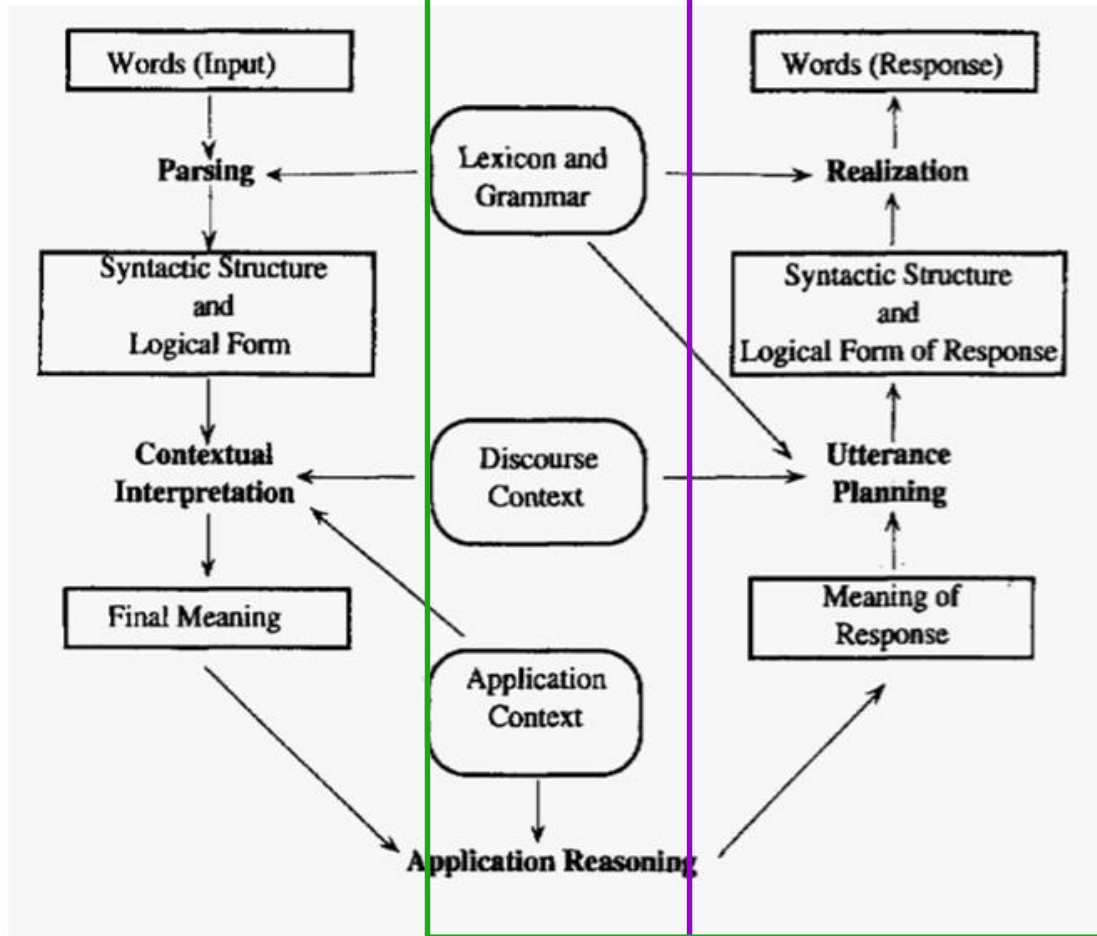
Level	Focus	Agentic AI Function
Syntactic	Grammar/Structure	Parsing user commands and API documentation.
Semantic	Meaning/Intent	Identifying the core task (e.g., "book flight").
Pragmatic	Context/Goal	Resolving ambiguity based on persona, history, and goals.

The Organization of Natural Language Processing Systems



Allen 1995: Natural Language Understanding - Introduction

**Natural
Language
Understanding**

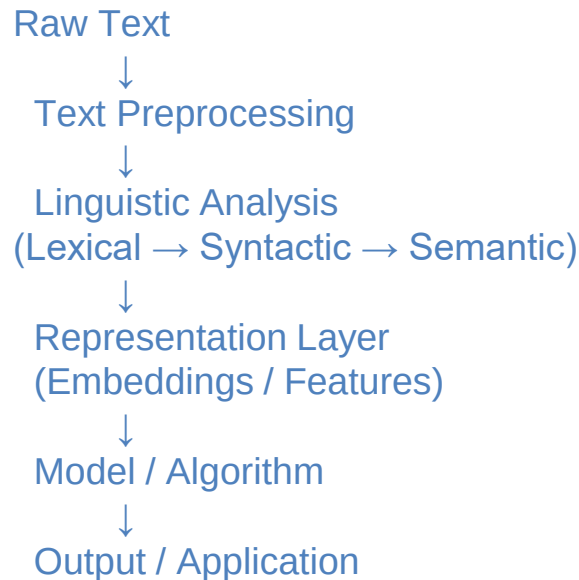


**Natural
Language
Generation**

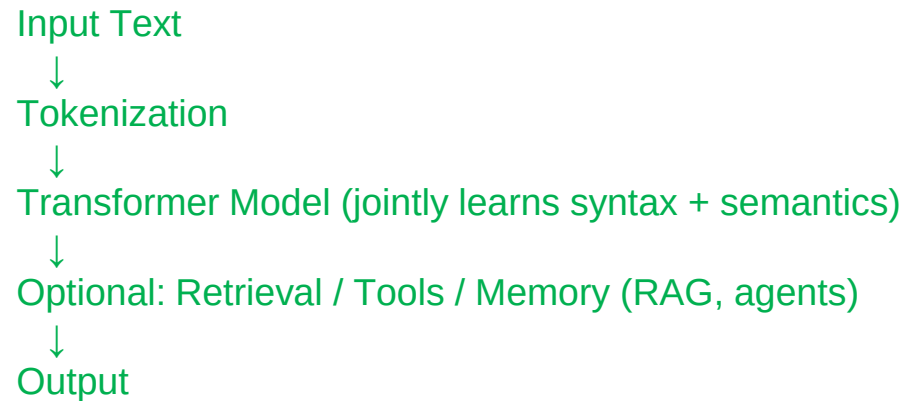
Classical, Modern and Hybrid NLP Systems



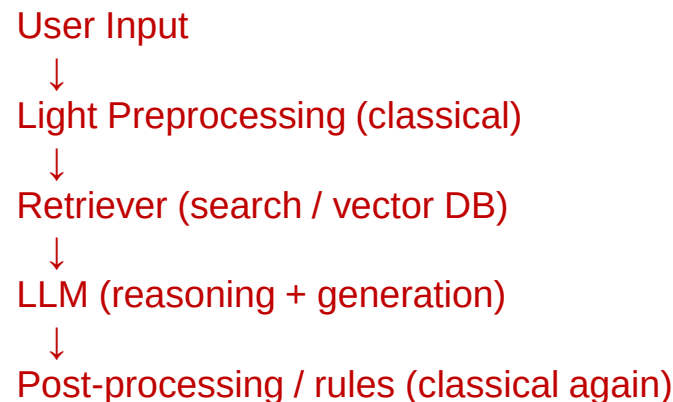
Classical NLP System



Modern NLP System



Hybrid NLP System



Evaluating NLP Systems

- What metrics to use?
- Is NLP systems reliable?
- Is the sample of texts sufficiently representative?
- How reliable or certain are the results?

Good NLP evaluation goes beyond accuracy —it measures reliability, fairness, and real-world usability.”

Evaluating NLP Systems



- **Task-specific Performance**

Use the right metrics (accuracy, F1, BLEU, ROUGE) based on the task.

- **Data & Generalization**

Evaluate on diverse, unseen datasets to ensure real-world performance.

- **Robustness & Fairness**

Handle noisy inputs and check for bias in outputs.

- **Efficiency & Scalability**

Measure latency, throughput, and cost for deployment.

- **Human Evaluation & Error Analysis**

Assess quality (fluency, relevance) and analyze failures for improvement

Evaluating NLP Systems

- **Accuracy / F1 Score**

Used for classification tasks (e.g., sentiment analysis, spam detection).

- **Precision & Recall**

Important when false positives/negatives matter (e.g., medical or fraud detection).

- **BLEU Score**

Measures quality of generated text against reference (commonly in machine translation).

- **ROUGE Score**

Evaluates overlap with reference summaries (used in text summarization).

- **Perplexity**

Measures how well a language model predicts text (lower is better).

Contingency Table

		gold standard labels		
		gold positive	gold negative	
system output labels	system positive	true positive	false positive	precision = $\frac{tp}{tp+fp}$
	system negative	false negative	true negative	
		recall = $\frac{tp}{tp+fn}$	accuracy = $\frac{tp+tn}{tp+fp+tn+fn}$	

Good References



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<https://analyticssteps.com/blogs/top-nlp-tools>

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https://www.cstr.ed.ac.uk/emasters/course/natural_lang.html
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Dr. Chetana is an Professor and Program lead in the CSIS department at Work Integrated Learning Programmes Division, BITS Pilani.

She has more than 28 years of teaching and industry experience. She did her PhD in Computer Science and Engineering from a joint programme of IIT Bombay and Monash University, Australia. She has been working extensively on different state of art research projects and has been awarded the “Best Industry Aligned Research” at the CSI TechNext India 2019 - Awards to Academia. She has published various papers and is also a reviewer at national and international level peer reviewed conferences and journals. Her areas of expertise include Machine Learning, Natural Language Processing, Semantic Web, Deep Learning, Text Mining, Big Data Analytics, Information Retrieval and Software Engineering.

Thank you!!